MATH 220/199 Practice Final Exam

Name:

Instructions

- Please write your name.
- Show as much work as you can for every question.
- Attempt every single question.
- Attempt every single question.
- I cannot emphasize this enough - Attempt every single question.
- The use of electronics - including a calculator - is prohibited. Leave your answers in exact form if you must.
- Any form of cheating is strictly prohibited.
- GOOD LUCK!!!

REMARK: This practice exam is not necessarily comprehensive, but every topic in this practice exam is fair game for the actual exam.

Chapter 1: Functions and Models

1. Let $f(x) = x^2 - \sqrt{2x}, g(x) = \cos(2x)$.
   (a) Compute $f(g(x))$
   (b) Compute $g(f(x))$

2. Given that $\sin(\theta) = \frac{3}{\sqrt{10}}$ Compute the following quantities:
   (a) $\cos(\theta)$
   (b) $\cot(\theta)$
   (c) $\sin(2\theta)$

Chapter 1: Functions and Models (Contd)

1. In the space below, make a sketch of $f(x) = 2\ln(x-2)$. Label any asymptotes, any x-intercepts, and one non-x-intercept point on the graph.

2. Let $g(x) = 3e^{5x-1} + 2$. Compute $g^{-1}(x)$. State the domain and range of $g^{-1}(x)$.

Chapter 2: Limits and Derivatives

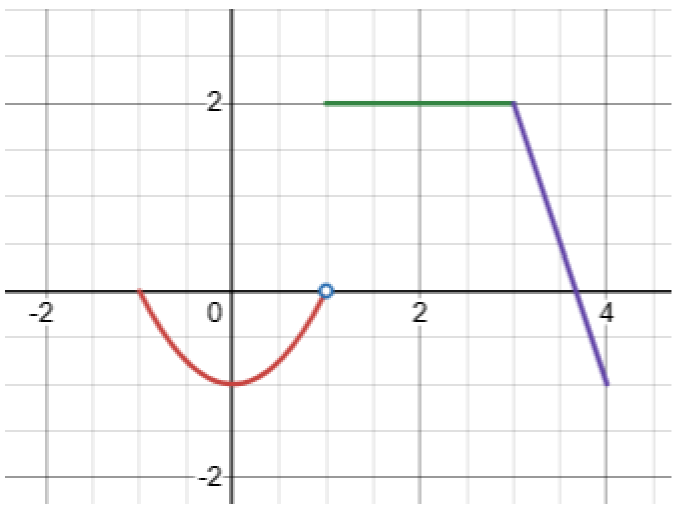
1. Compute $\lim\limits_{x\to 9} \frac{x-9}{\sqrt{x}-3}$

2. Compute $\lim\limits_{x\to -\infty} \frac{\sqrt{2x^2+1}}{4x}$

Chapter 2: Limits and Derivatives (Contd)

1. Given $f(x) = 3x^2 + 2x$ Use the limit definition of the derivative to compute $f'(x)$.

2. Consider $f(x)$ which is sketched below:



Compute the following or write DNE if the value does not exist:

(a) $f(1)$

(b) $\lim_{x \to 1} f(x)$

(c) $\lim_{x \to 3^-} f(x)$

(d) $f(2)$

Chapter 3: Differentiation Rules

For each of the following functions, compute the derivative:

1. $f(x) = \sin^{-1}(e^{5x^2+5x})$

2. $g(x) = \frac{\ln(2x)}{5^x}$

3. $h(x) = (\sqrt{x})^3 \cdot \tan(x)$

Chapter 3: Differentiation Rules (Contd)

1. A cylinder with constant height 10 ft. has a radius which is increasing at a rate of 2 ft/min. When the volume of the cylinder is 40π ft^3, how fast is the surface area of the cylinder increasing?

2. Using an appropriate $f(x)$ and the Method of Linearization, find an approximation of $(2.01)^3$.

Chapter 4: Applications of Differentiation

1. Let $f(x) = \frac{1}{x} + 4x^2$
   (a) Compute all the asymptotes of $f(x)$.
   (b) Compute the x-intercepts and y-intercepts of $f(x)$.
   (c) Compute the critical points of $f(x)$.
   (d) Use (a), (b), and (c) to make a sketch of $f(x)$ in the space below.

Chapter 4: Applications of Differentiation (Contd)

1. Compute the following limit: $\lim_{x \to \infty} \frac{3x^2+7x}{e^{2x}}$

2. Prove the following statement: In the interval $(-\frac{\pi}{4}, \frac{3\pi}{4})$, the function $g(x) = \sin(x) + \cos(x)$ has a critical point.

3. A fence for feral koalas is being built, and it must have an enclosed area of 400 ft^2. The north and south sides of the fence are being built with material which costs 20 dollars per foot, and the east and west sides are being built with material which costs 10 dollars per foot. Find the dimensions of the fence which minimizes the total cost.

Chapter 5: Integrals

1. Given: $f(x)$ is an odd function and $\int\limits_{-2}^{7} f(x)\ \mathrm{d}x = 32$, compute $\int\limits_{7}^{2} f(x)\ \mathrm{d}x$.

2. Compute $\frac{d}{dx} \int\limits_{31}^{x^2-x} \ln(3t^2+1)\ \mathrm{d}t$

3. Compute $\int\limits_{1}^{4} 4x\sqrt{x^2+1}\ \mathrm{d}x$

Chapter 5: Integrals (Contd)

1. An ultra-marathoner runs at a rate given by $v(t) = 3t^2 - t + 2$ (in miles per hour). At hour 2, the runner is already 10 miles into the course. How far will the runner be at hour 9?

2. Use the limit definition of the integral to compute $\int_{-1}^{3} 3x - 4 \, \mathrm{d}x$. (HINT: You can use usual integration to check your answer).

Chapter 6: Applications of Integration

1. Let R be the region bounded between $f(x) = x^2 + x$ and $g(x) = 2x + 2$. Compute the area of R.

2. Let K be the region enclosed by $f(x) = \sqrt{x-2}, x = 5, y = 0$.

   (a) Setup but do not evaluate an integral to find the volume of the figure obtained by rotating K around the y-axis.

   (b) Setup but do not evaluate an integral to find the volume of the figure obtained by rotating K around the x-axis.

Challenge Problems

1. Determine the set of all $n \in \mathbb{Z}$ for which $\int\limits_0^\pi \sin(nx)\ \mathrm{d}x = 0$.

2. Let $f(x) = \sin(-x) + e^{x\sqrt[3]{3}} + \ln(x\sqrt[7]{7})$. Let $f^{(n)}$ denote the n^{th} derivative of $f(x)$. What is the smallest n for which $f^{(n)}$ has integer coefficients for every term?

Page for Scratch Work + Remarks

Remarks

The following topics were not covered on this practice exam but are fair game for the final. You may want to revisit worksheets or quizzes which covered the following topics/techniques:

1. Chapter 1:
   (a) Trigonometry
2. Chapter 2:
   (a) Continuity
3. Chapter 3:
   (a) Expect harder material on the final regarding related rates, trigonometric derivatives, logarithmic derivatives, and linearization.
4. Chapter 4:
   (a) Curve Sketching
   (b) Anti-derivatives
5. Chapter 5:
   (a) Make sure to review Exam 3 when it is published to see what you missed.
6. Chapter 6:
   (a) Make sure to review Exam 3 when it is published to see what you missed.
   (b) More comprehensive practice which covers more cases can be found here: https://tutorial.math.lamar.edu/Problems/CalcI/VolumeWithRings.aspx

Space for Scratch Work